

Data Analysis and Insights Brief

1. Framing

The goal is to examine user and critic scores of videogames over an 8-year period (2011-2019) and extract insights into differences in how these groups consume and evaluate videogames, and how cultural shifts impact scores.

Key metrics:

Genre, Rating, Release Date, User Scores, Critic Scores

Cohort Analysis between User and Critic Data

Data Overview and Limitations:

The dataset was retrieved from Kaggle.com and includes data collected from Metacritic for over 5500 videogames released between 2011 and 2019. This dataset is not necessarily representative of all videogames released during this time period (e.g., over 1000 games can be released in a single year) and not all released games will have data represented in metacritic.com.

Additionally, user scores on Metacritic are susceptible to 'review bombing,' where users deliberately leave low scores in response to controversies unrelated to game quality. This may artificially deflate user scores for certain high-profile titles and should be considered when interpreting user score trends over time. There is also no qualitative date (i.e., reviews) that can be used to assess specific points of praise/criticism that may shift over time with cultural changes in the videogame industry.

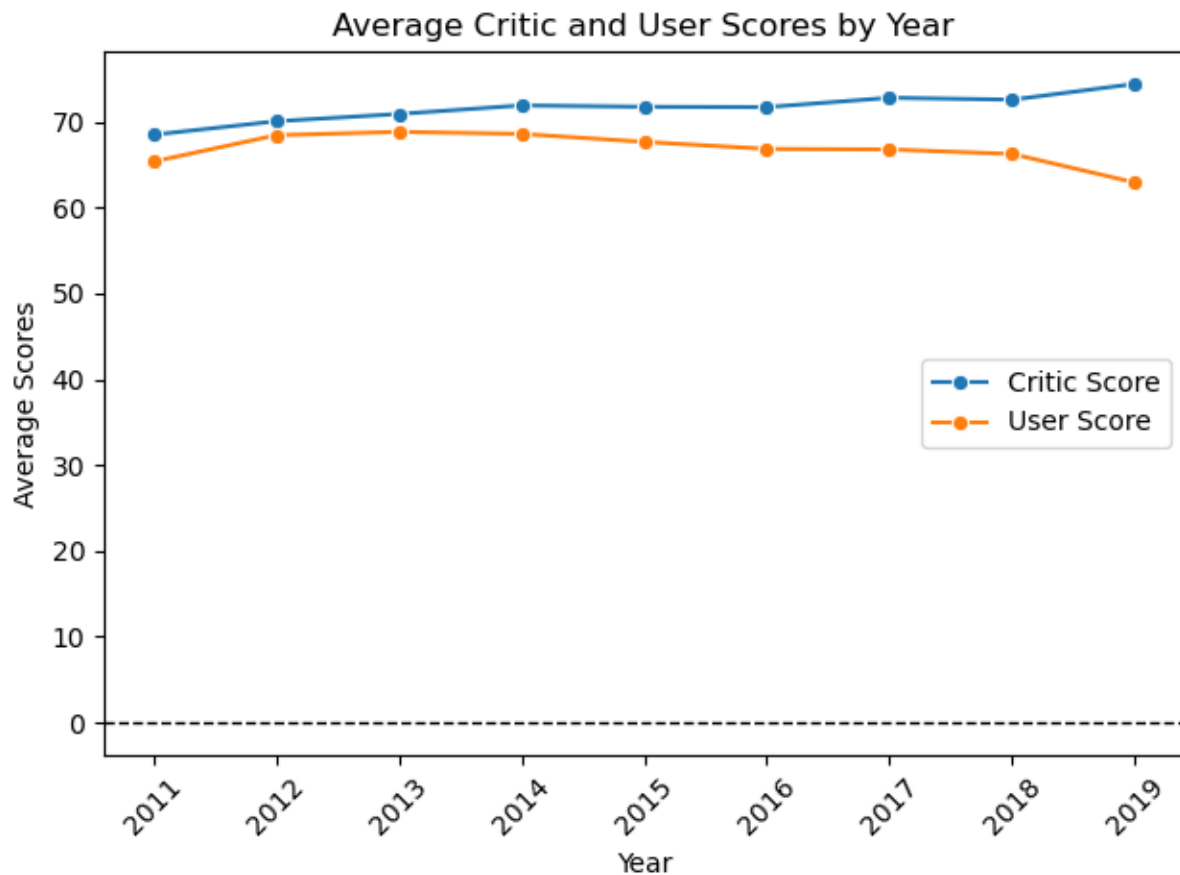
2. Data Cleaning Steps

The majority of data was complete with no missing values, particularly when examining the critic scores and user scores per game. Initial analyses of genre data revealed several categories that were only represented by 1-2 titles. These more unique genres appeared more often in analyses due to a lack of variance in data. Games with more unique genres were assessed and reassigned more general genre names in order to be better represented in data.

For example, one game out of 5700 was given the genre of “WWI” and was assigned a more general genre of “Simulation”.

3. Visualizations

Figure 1:

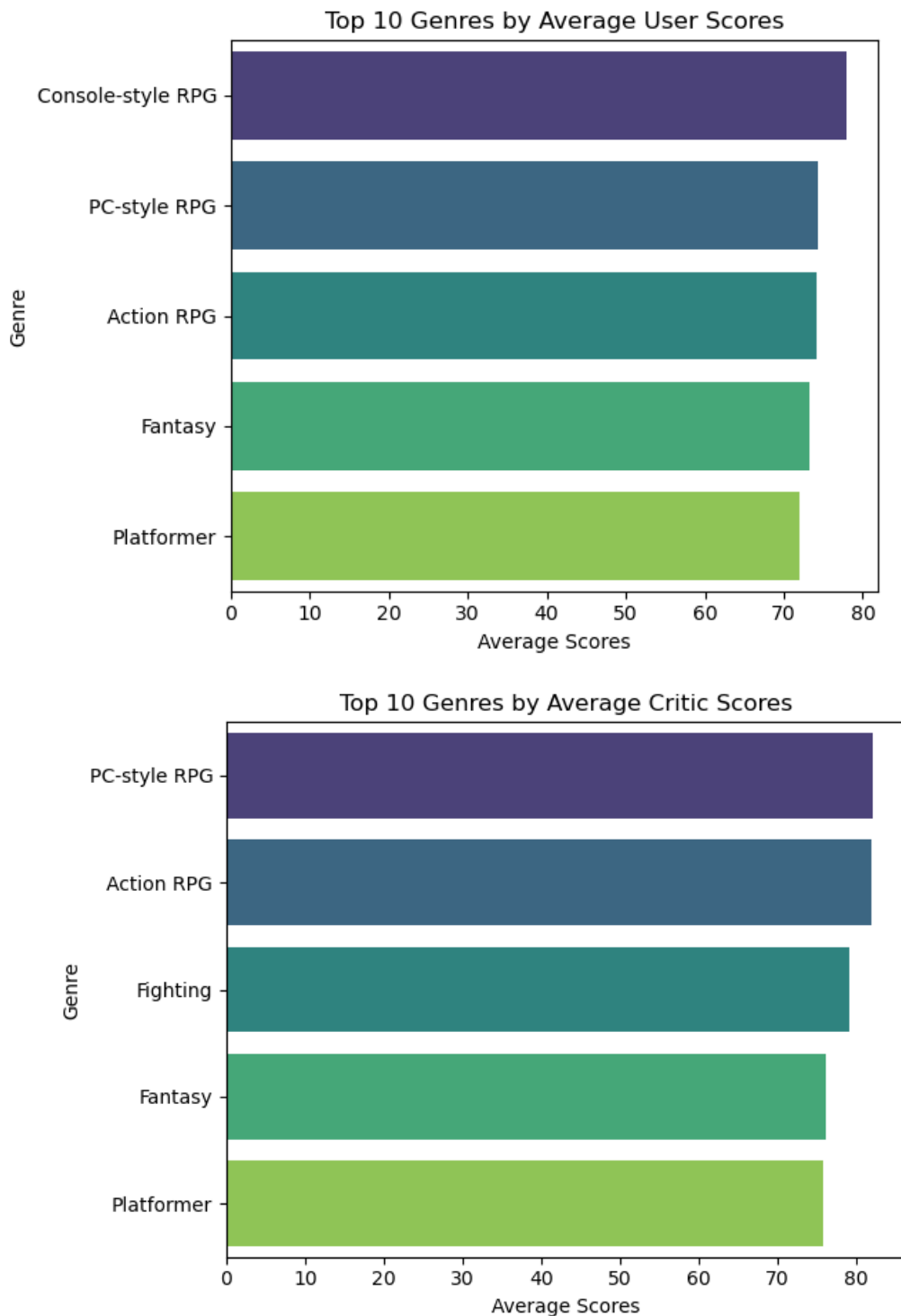


Examining average critic and user scores over time reveals that these scores were nearly identical in 2011 and gradually began to diverge more over the years, with average critic scores increasing and average user scores decreasing. A few reasons for this divergence may include:

- Changes in game development practices, such as the prevalence of microtransactions and always-online requirements for some offline games, caused users to give harsher scores to games that followed these practices. By contrast, these issues didn't factor as much into critic scores.
- Earlier-released games typically only received reviews from more engaged plays who enjoyed the experience. As Metacritic became more well-known, more casual players

began leaving more reviews and they were more likely to share feedback with games they did not enjoy.

- As social media became more prevalent in the early 2010s, users became more aware of scandals and controversies in the videogame industry, sometimes with regard to specific developers and business practices. Metacritic soon became a place to “protest” games by giving low scores rather than giving more objective feedback. Critics are expected to evaluate games based on their own merits rather than corporate ethics.
- [Figure 2:](#)



These figures show very similar rankings of the top-rated genres between users and critics, suggesting there is a strong level of agreement over what genres tend to be “good”. However, there are a few notable differences:

- Console-style RPGs tend to take a long time to finish (40-80 hours), which is a time commitment that critics are not able to meet in order to thoroughly review the game. Users, on the other hand, have much more time to invest and are more likely to leave positive reviews of this genre. They also enjoy more narrative-focused games that have clear character development and a satisfying conclusion.
- Conversely, fighting games have higher critic scores as they are easy to pick up and play and evaluate quickly. The core gameplay doesn't change much and doesn't require a significant time investment. Critics also tend to be more familiar with the mechanics of fighting games. Casual users, on the other hand, may be turned off by the highly technical nature and steep learning curve of fighting games and more likely to leave a negative review.
- Overall, data seem to suggest that users value quality of experience while critics value quality of design.

4. Recommendations

For developers:

- Focus on user-experience for genres like console-style RPGs by prioritizing strong character arcs, emotional story beats/environments, and satisfying endings.
- Focus on enhancing technical aspects if you want a critically-acclaimed game, such as balanced game mechanics, systemic depth, and clear design intentions
- Be careful with post-launch changes and increased monetization as these can lead to increased criticism from users and risks alienating your audience.

For consumers:

- Interpret critic scores differently based on the genre and remember that critics do not have the time to invest heavily into story-driven games.
- Read user scores and look for common themes of praise or criticism in order to better identify “protest” reviews and lead to more objective insights.
- Look for reviews that emphasize the aspects you care about most, such as time investments, mechanical features, story quality, etc.

5. Conclusion

This analysis reveals that while critics and users broadly agree on which genres produce quality games, meaningful differences exist in how each group evaluates individual titles and genres. This divergence reflects structural differences in how critics and users engage with games, as well as broader cultural shifts in the gaming industry between 2011 and 2019. Understanding these differences is valuable for developers, publishers, and consumers to interpret review scores more critically and make more informed decisions.